A Typed Claim-Governance Ontology for Neuroscience and SAN Research

Author: Micah Blumberg

Self Aware Networks Research Institute

https://selfawarenetworks.com

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Abstract

Knowledge graphs can make research claims easier to retrieve while also making unsupported claims easier to repeat. This paper presents an executable, provenance-aware claim ontology designed for a mixed corpus containing conventional neuroscience results, public standards, private design notes, and unvalidated Self-Aware Networks (SAN) proposals. The revision replaces a v1 JSON smoke fixture with a Draft 2020-12 JSON Schema, 24 typed claim cases, 12 source identities, 15 relation classes, ten exact-set competency questions, and mutation-tested policy code. Public citation eligibility is defined as a conjunction of privacy, evidence class, claim status, edge verdict, source class, allowed action, and blocked action. A privacy-only baseline and the governed policy are evaluated on 16 development cases and eight held-out cases. The baseline obtains precision 0.444444, recall 1.0, and F1 0.615385 overall; on the held-out partition it obtains precision 0.571429, recall 1.0, and F1 0.727273. The governed policy exactly matches the 24 author-coded labels. This perfect agreement is an implementation result, not a scientific result: the corpus is synthetic, the labels were authored with the policy, no independent annotation was performed, and no reliability estimate exists. The artifact therefore demonstrates reproducible claim gating on its declared cases. It does not validate SAN, phase-wave differentials, Neural Array Projection Oscillation Tomography (NAPOT), consciousness, priority, or a biological mechanism.

Scope and evidence boundary

Successor manuscript for Zenodo record 21231148 under its existing concept DOI.

1. Problem and contribution

Suppose a reader asks a research graph for evidence supporting a claim. The graph may return a nearby paper, a private design note, an author's own hypothesis, or a relation that an editor already rejected. All four items can be retrievable while only one may be appropriate to cite as established background. The practical problem is therefore not merely storing claims; it is preventing retrieval from silently becoming endorsement.

A solution needs to record what kind of claim each item is, where it came from, what judgment was made about it, whether it is public, and what a downstream system may do with it. In this paper, an ontology that encodes those distinctions and tests the resulting action rules is called a **typed claim-governance ontology**.

A graph edge has at least three meanings that must not be collapsed. It can record that a source made a statement, represent that an editor accepts a relation for a bounded purpose, or encode a hypothesis to be tested. A graph that stores these meanings in the same undifferentiated relation invites a category error: retrieval is mistaken for evidential support. Vector similarity and graph proximity create a related problem. They can help locate candidate material, but high rank is not a scientific warrant.

The v1 paper recognized this risk in prose and attached source, verdict, privacy, and action fields to a small JSON fixture. Its implementation did not support the validation language used around it. Parsing two nodes and five edges established machine readability, not ontology validity. Four query descriptions were not executable queries. Four gold rows did not constitute a documented benchmark, and the source ledger repeated one local path in place of the retired Downloads copy.

This revision makes a narrower contribution. It specifies and tests an executable policy for deciding whether a claim in a heterogeneous research graph is eligible for public citation. The contribution has four parts:

1. a typed claim and source model with relation-specific requirements;
2. a deterministic public-citation rule and an intentionally weak baseline;
3. exact-set competency questions and adversarial mutation tests; and
4. explicit separation of artifact conformance from scientific validation.

The ontology is not presented as a comprehensive neuroscience ontology. It is a governance layer for a particular research-development corpus.

2. Provenance correction and source roles

The canonical design source is the CORE copy of *Neuroscience Ontology Analysis*. The retired source was a Downloads copy that is no longer present. The v1 manuscript accidentally printed the CORE locator for both rows. The current dependency ledger restores the two identities and hashes the canonical file. This is a provenance correction, not evidence for a neuroscientific claim.

The content of that canonical file is also reclassified. It begins as a model-assisted assessment and continues as a design exchange. It can document why relation classes and policy controls were proposed. It cannot serve as an independent ontology study, empirical result, or external validation of SAN. The current private SAN manuscript and the private rank-17 comparison ledger have the same evidential limitation here: they provide definition continuity and internal consistency checks, not independent support.

External sources serve bounded background roles. Communication through coherence concerns phase-dependent effective communication between neural populations [1]. Reviews of cortical traveling waves establish that propagating activity patterns have studied mechanisms and computational roles [2]. Monkey visual-cortex recordings distinguish alpha- and gamma-associated feedback and feedforward processing [3], while working-memory studies report brief gamma and beta burst structure [4]. These sources justify background claim examples; they do not establish SAN-specific constructs.

The standards sources define machine-governance concepts. PROV-O supplies an OWL 2 provenance ontology [5], PROV-DM defines a provenance data model [6], SHACL defines shapes and constraints for validating RDF data graphs [7], and the FAIR principles motivate rich metadata, qualified references, licensing, provenance, and community standards for reuse [8]. The present JSON artifact does not claim to be an RDF, OWL, or SHACL implementation. These standards are comparison anchors for a future interoperable representation.

3. Claim model

Each claim instance contains a stable identifier, subject, relation, object, statement, relation-specific attributes, evidence class, claim status, edge verdict, privacy tag, source identifiers, source anchor, allowed actions, blocked actions, benchmark split, and author-coded public-citation label.

Evidence class identifies the kind of source rather than the truth of the statement. The implemented classes are external primary research, external review, public standard, author public source, private design source, and model proposal. A public author source may support what the author previously wrote, but it does not become independent scientific confirmation. A private design source can preserve provenance while remaining ineligible for public citation.

Claim status records editorial interpretation. established_background is reserved for bounded statements supported by an eligible external or standards source. `source_summary` reports what a source says without accepting the statement as established. `conjecture` marks an unvalidated hypothesis. implementation_proposal marks a model or software target. `rejected` preserves an excluded relation so that the system can warn against its reintroduction.

Edge verdict is a separate field: accepted, deferred, rejected, or blocked. Status and verdict answer different questions. A source summary may be accurately transcribed but deferred for public use. A conjecture can remain a legitimate research target without being accepted as established background. A rejected row can remain queryable as negative knowledge.

Privacy tags distinguish public-safe, private-source, and local-only content. Action fields then make the operational consequence explicit. Typical actions include query, compare, cite, or retain for provenance. Blocks include publicize, use as proof, and promote without review. This design prevents a single `public_safe` flag from carrying the full burden of evidence review.

4. Relation dictionary

The artifact implements 15 relations. Conventional graph relations include `BrainRegion` and `Connection`. Dynamic neuroscience relations include `Oscillation`, `PhaseCoupling`, `NeurotransmitterEffect`, and `GatingRelation`. Provenance and validation relations include `ProvenanceLink` and `ValidationShape`. Research-program relations include `PhaseWaveDifferential`, `TomographicProjection`, `CoincidenceEvent`, `ScaleBridge`, `GlobalSpectralMap`, and `AgenticUnit`. `AgentMemoryUse` records how a retrieval system may use an item.

Each relation declares required attributes. A phase-coupling claim must state the source and target, frequency context, direction source, delay or offset, and coupling measure. A phase-wave-differential claim must distinguish the measured phase terms, wrapping convention, propagation delay, receiver frame, and independently supplied direction. Direction must be supplied independently; phase difference alone does not determine a directed causal route.

A tomographic-projection claim must identify its projection family, reconstruction target, identifiability assumptions, redundancy condition, and validation plan. The NAPOT row therefore remains a conjecture about what would need to be demonstrated, not a conclusion drawn from observing traveling waves. A scale bridge must name its microstate, macrostate, order parameter, scaling rule, and falsification condition. A coincidence event must specify a receiver, timing window, threshold rule, output observable, and population context; it is not automatically one spike or one bit.

Tonic and phasic are represented as relative functional roles, not universal synonyms for fixed low- and high-frequency bands. The global-spectral-map row is a nested multiscale phase-field proposal rather than a unitary broadcast. The agentic-unit row is blocked from promoting every neuron to a phenomenally conscious subject. There is no central spectator or literal screen in the rendering metaphor. These boundaries make the SAN concepts testable without silently converting metaphors into anatomy or established mechanism.

5. Citation policy

For claim c with source set S(c), public citation eligibility is:

```
eligible(c) = public_safe(c)
              AND established_background(c)
              AND accepted(c)
              AND all source_class(s) in E for s in S(c)
              AND cite is allowed
              AND publicize is not blocked
              AND use_as_proof is not blocked,
```

where E contains external primary research, external review, and public standards. This is a deliberately conservative rule for the present corpus. It is not asserted to be a universal scholarly-communication policy.

The baseline predicts eligibility whenever the privacy tag is `public_safe`. It captures a common failure mode: content is treated as publishable because it is not secret. That rule ignores whether the claim is a conjecture, implementation proposal, rejected relation, or source summary and whether its sources are independently eligible.

The governed rule is deterministic. It has no trained parameters, embedding model, probabilistic threshold, or hidden prompt. Its purpose is auditability. The code returns the same result from the same versioned data and dependency set.

6. Benchmark construction

The benchmark contains 24 hand-authored claim cases. Eight positive cases are bounded background or standards statements. Sixteen negative cases include private summaries, SAN conjectures, implementation proposals, rejected claims, and local provenance operations. Sixteen cases are assigned to development and eight to held-out. The held-out partition contains four positive and four negative gold labels.

The term held-out requires qualification. The cases were not annotated by a blind external team after the policy was frozen. They were designed and labeled as part of the same artifact. The split detects accidental dependence on the development-case identifiers and exercises the same rules on unused fixture rows; it does not estimate out-of-distribution generalization.

Ten competency questions specify exact expected identifier sets. They ask for all and held-out public-citable claims, conjectures, rejected claims, blocked claims, selected relation types, selected source classes, and development-only eligible claims. Exact-set comparison tests both false additions and omissions.

JSON Schema Draft 2020-12 validates structure and enumerated values. Additional semantic checks enforce unique claim and source identifiers, the 16/8 split, relation-specific attributes, valid source references, agreement between gold labels and the declared policy, and mandatory blocks for rejected, private, conjectural, and proposed claims. Source checks prohibit redistribution and keep independent review and public action false.

Fifteen mutation tests challenge missing fields, invalid enums, duplicate IDs, unknown sources, missing relation attributes, private-claim promotion, rejected-claim promotion, conjecture promotion, gold-policy mismatch, ineligible sources, wrong expected query sets, unknown query types, source redistribution, and public authorization. These tests target the fail-closed behavior of the package.

7. Results

All five benchmark checks pass: schema, source manifest, semantic invariants, competency questions, and metrics. All ten exact-set competency questions pass, and all 15 mutation tests pass.

Policy and partition	TP	FP	FN	TN	Precision	Recall	F1
Privacy-only, all 24	8	10	0	6	0.444444	1.000000	0.615385
Governed, all 24	8	0	0	16	1.000000	1.000000	1.000000
Privacy-only, held-out 8	4	3	0	1	0.571429	1.000000	0.727273
Governed, held-out 8	4	0	0	4	1.000000	1.000000	1.000000

The ten overall baseline false positives are the intended demonstration. A privacy-only rule admits public-safe conjectures, proposals, and rejected claims. The governed rule blocks them because evidence, status, verdict, source, or action conditions fail.

These values are descriptive statistics for a constructed fixture. They have no sampling uncertainty because no population sample was drawn, and no confidence interval would turn them into a biological performance estimate. The perfect governed result is largely a consistency check: the gold labels encode the same declared policy. It should not be advertised as classifier accuracy on an independent corpus.

8. What the result does and does not establish

The result establishes that the artifact is executable, deterministic, and fail-closed for the tested mutations. It establishes that the public-citation query returns exactly the authored eligible set. It also exposes why privacy alone is insufficient in this corpus.

The result does not establish ontology completeness. Fifteen relations do not cover neuroscience. It does not establish annotation validity or reliability. It does not show that another editor would assign the same evidence classes, statuses, verdicts, or gold labels. It does not establish a benefit over SHACL, OWL, PROV-O, or another knowledge-graph technology. It does not test graph scalability, natural-language extraction, entity resolution, or adversarial prompt behavior.

Most importantly, graph membership is not SAN validation. PWD, NAPOT, the global spectral map, coincidence events, scale bridges, and agentic units are represented so that their requirements and prohibitions can be inspected. That representation supplies no empirical support by itself. Compatibility with a neuroscience result is likewise not independent validation of SAN. The ledger is an editorial source map, not a neuroscience experiment.

9. Provenance, rights, and release controls

The source manifest stores locators and source classes, not third-party source bodies. Peer-reviewed articles and standards are cited or paraphrased within ordinary scholarly bounds and are not redistributed. Private design notes and private manuscript dependencies are excluded from the public release. The versioned manuscript, synthetic benchmark data, schema, validator, tests, and generated result are released together under CC BY 4.0 so that the bounded implementation claim can be reproduced.

The package pins the v1 manuscript, v1 fixture, published PDF, canonical design source, current SAN definition handoff, rank-17 overlap ledger and result, and the provenance-paper manuscript by byte count and SHA-256. The missing retired Downloads path is recorded as expected absent. This preserves source identity without pretending that a missing duplicate changes the substantive record.

This successor remains under the existing concept DOI and identifies the v1 source-path correction. The local final-preprint package freezes the title, benchmark endpoint, relation dictionary, labels and split, SAN conjecture boundaries, public source scope, CC BY 4.0 license, and exact artifact set. Uploading that package or changing provider metadata remains a separate administrative action.

10. Next validation stage

The present final preprint reports the completed bounded implementation study. Its next scientific validation stage requires independent work rather than more internally consistent labels. Ontology and neuroscience reviewers should audit the relation dictionary and background claims. Two or more independent annotators should label a frozen, larger claim set without seeing the policy outputs; disagreements and an appropriate reliability statistic should be reported. The gold adjudication protocol must be written before that external evaluation.

The enlarged corpus should include difficult negatives, source conflicts, partially supported claims, mixed public / private provenance, retractions or version changes, and multi-source claims where one source is ineligible. A frozen external test set should be evaluated only after relation definitions, eligibility policy, and thresholds are locked. Baselines should include more than privacy, such as status-only and source-class-only rules. Error analysis should report which gate prevented each false promotion and whether a valid claim was overblocked.

An interoperability study can then map the JSON model to RDF, PROV-O, and SHACL shapes. That study should test round-trip preservation of claim status, privacy, allowed actions, blocked actions, and source identity. It should not claim superiority without a comparative task and predefined metrics.

11. Conclusion

The corrected paper offers a modest but reproducible result. A heterogeneous neuroscience / SAN graph can encode source, evidence, status, verdict, privacy, and action constraints as executable policy. On a 24-case synthetic benchmark, that policy blocks ten false promotions made by a privacy-only baseline and passes all authored competency and mutation tests.

The scientific boundary is equally important. The benchmark measures contract behavior on author-coded cases. It does not measure neuroscience truth, validate SAN, or convert a relation into evidence. The next meaningful advance is independent annotation and review on a frozen external corpus, not stronger language about the perfect internal score.

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